CSE322: Introduction to Embedded Systems

Fall 2026

Project Description

Home Appliances Control

Each team of **4-6 members** builds a PC-controlled home-appliance prototype on the TM4C123GH6PM board, using a PuTTY terminal over the board's USB cable, and submits code, report and demo video by Week 13.

Project Overview

The prototype lets a PC control and monitor a small home through the TM4C123GH6PM board. You type commands in PuTTY (serial, 115200 baud) and the board replies. The board must do the following:

- Switch a 220 V lamp on and off with a command. Wire the relay contacts in parallel with the existing manual switch, so the manual switch keeps working.
- Switch a power plug on and off with a command. When the plug relay is off, the plug has no power.
- Report the door status (open or closed), read from a magnetic reed switch, every time it changes.
- Send the room temperature to the terminal every second, read from an LM35 sensor.
- Let the user set a temperature limit with a command. Above the limit, the board prints an alarm line.
- Keep the time of every door opening and closing and print the list on request.

Hardware Requirements

Each team needs these components:

- One TM4C123GH6PM LaunchPad board
- A PC with PuTTY and the LaunchPad's USB cable
- One LM35 temperature sensor
- One magnetic reed switch and its magnet, for the door (or a pushbutton)
- One 5 V relay module with 2 channels, one for the lamp and one for the plug
- One external 5 V power supply (at least 1 A) for the relays and the LM35
- A 220 V lamp with its manual switch, and a power plug
- A panel to mount everything on, with the wires hidden

Pin map

Every team uses the same pins, so one test setup can check all teams. Power the relays and the LM35 from one external 5 V power supply. Connect the supply's GND to a GND pin of the LaunchPad so that all grounds are common. The LaunchPad itself is powered by its USB cable, and the same cable carries the terminal link to the PC (UART0 on PA0 and PA1, through the on-board debugger: no extra wires).

Component	Module pin	TM4C123 pin
Lamp relay module	IN	PA2
Plug relay module	IN	PA3
Door reed switch	one leg (other leg to GND)	PA4
LM35 temperature sensor	OUT	PE3 (AIN0)
USB virtual COM port (to PuTTY)	on the LaunchPad	PA0 (U0Rx)
USB virtual COM port (to PuTTY)	on the LaunchPad	PA1 (U0Tx)

Firmware layers (in the firmware folder of your project):

- MCAL: your drivers, one module per peripheral (for example dio.c/.h, uart.c/.h, adc.c/.h, systick.c/.h), written on the registers or on TivaWare.
- HAL: one module per device, each with its own .h and .c: lamp, plug, door, lm35, serial (UART0) and timebase. The starter project gives the six .h files; do not change their function signatures. Each .c uses your MCAL.
- Application: app.c and app.h with the terminal protocol, the alarm and the door log. It may include only the six HAL headers and standard C, so it runs unchanged on a PC for the automatic tests.

Terminal Protocol (PuTTY)

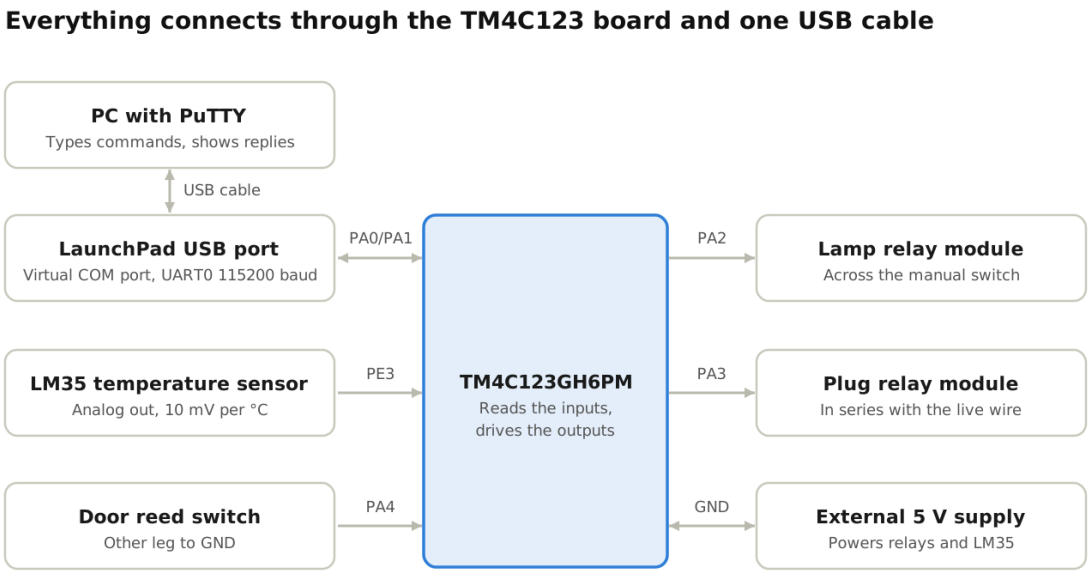
Open PuTTY with Connection type Serial, the LaunchPad's COM port (Device Manager, Ports), speed 115200, 8 data bits, no parity, 1 stop bit, no flow control. Everything is ASCII text and a command is one line. PuTTY sends CR when you press Enter, so the board accepts CR, LF or CRLF as the end of a line.

Direction	Message	Meaning
PC to board	LAMP ON or LAMP OFF	Lamp relay on or off. The board repeats the command as its reply, e.g. LAMP ON
PC to board	PLUG ON or PLUG OFF	Plug relay on or off. The board repeats the command as its reply
PC to board	LIMIT 30	Set the temperature limit to 30 °C (any whole number). The board replies LIMIT 30
PC to board	STATUS	Five lines: LAMP ON|OFF, PLUG ON|OFF, DOOR OPEN|CLOSED, TEMP 27.5, ALARM ON|OFF
PC to board	LOG	The last 5 door events, oldest first, one per line: LOG 12 OPEN (seconds since boot). LOG EMPTY if none
PC to board	HELP	Lists the commands
Board to PC	TEMP 27.5	Room temperature in °C, sent every second
Board to PC	DOOR OPEN or DOOR CLOSED	Sent on every door change
Board to PC	ALARM ON or ALARM OFF	ON when the temperature is above the limit, OFF when it returns to the limit or below. Sent once per change
Board to PC	ERR ...	Reply to an unknown or badly formed command (a line starting with ERR). Nothing changes

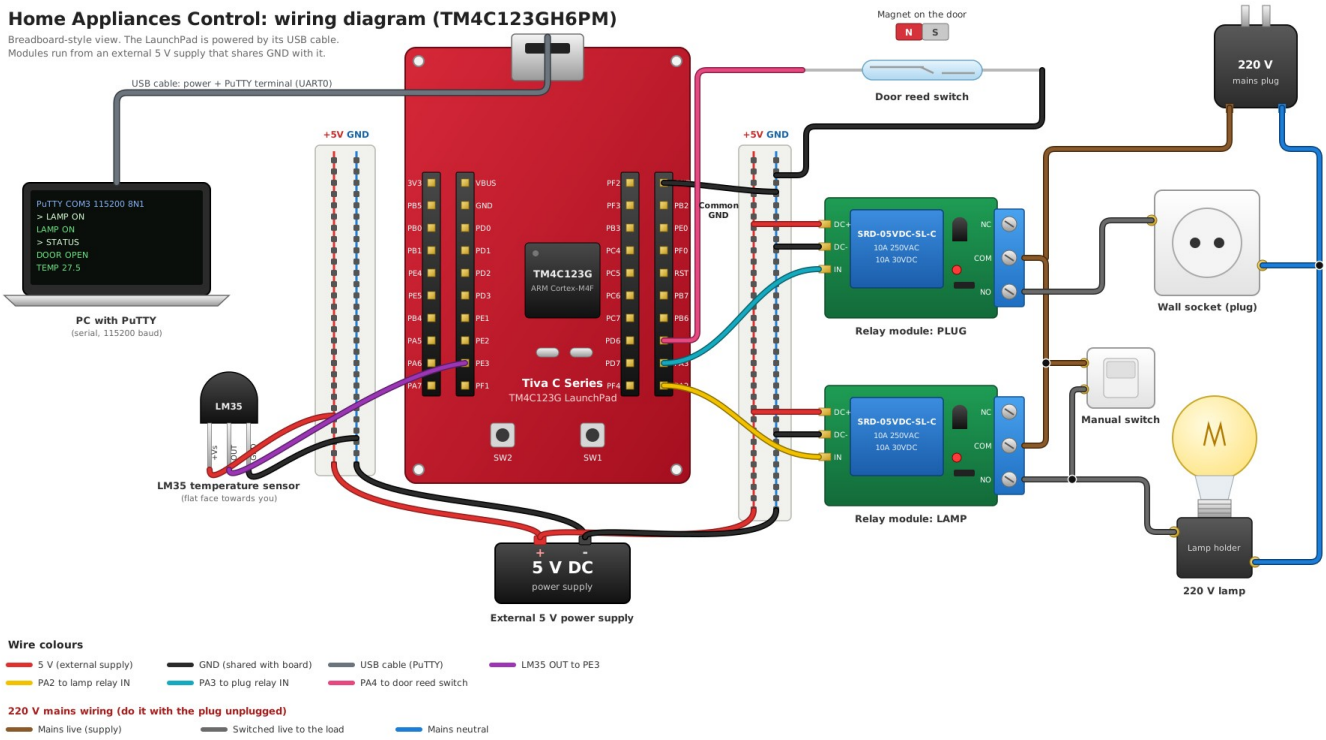
Terminal rules:

- Commands are not case sensitive, extra spaces are ignored, and a blank line does nothing.
- The board echoes every character you type, so you can see it in PuTTY, and handles Backspace: it removes the last character and erases it on the screen.
- Every line the board sends ends with CR LF, so PuTTY shows it correctly.
- A very long line must not crash the board or stop later commands from working.

Connection Diagram



Wiring Diagram



Project Deliverables

Each team submits one compressed file by the end of Week 13, containing:

1. The project folder: the TM4C123 firmware source (following the pin map and terminal protocol above).
2. A demo video of 5 minutes showing every requirement working on the real prototype.
3. A technical report in one PDF file with:
    - The contribution of each team member.
    - Flowcharts of the main program flow.
    - The list of components and the wiring diagrams.
    - Photos of the prototype.
    - Problems faced and how the team solved them.

Grading Rubric

Criteria	Marks	Details
1) Functional Requirements	**15**	
– Lamp control	2	LAMP ON/OFF switches the lamp relay and the board confirms it; the manual switch keeps working
– Plug control	2	PLUG ON/OFF switches the plug relay and the board confirms it; the plug has no power when off
– Door status	2	Reed switch read on the board; DOOR OPEN / DOOR CLOSED sent on every change
– Temperature display	2	LM35 read through the ADC; a TEMP line sent every second
– Temperature alarm	2	LIMIT sets the limit; ALARM ON / ALARM OFF sent once per change
– Door event log	2	LOG lists the last 5 door events with their times
– Terminal interface	2	Echo, backspace, CR / LF / CRLF, upper or lower case, ERR for bad input, long lines survive
– HELP and STATUS	1	HELP lists the commands; STATUS prints lamp, plug, door, temperature and alarm
2) Software Design & Report	**5**	
– Layered Architecture	1	Clear separation into MCAL, HAL and Application layers; the app uses only the HAL headers
– Hardware drivers	2	MCAL modules (GPIO, ADC, UART0, SysTick) used by one HAL module per device; the firmware compiles for the TM4C123
– Final Report	2	All required sections and pictures present
Total	**20**	

The marks come from automatic checks: protocol tests on a simulated board, static checks of the MCAL and HAL, and a structure check of the report. They cannot see your wiring or pin numbers, so test on the real board and show it in your demo video.